

William Huang

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EDUCATION

McMaster University

Bachelor of Engineering (Engineering 1)

Hamilton, ON

Sep. 2026 – Expected Jun. 2031

LEADERSHIP EXPERIENCE

Design Lead

2025 – 2026

FIRST Robotics Competition Team 1360

Oakville, ON

- Led a 35+ member mechanical sub-team in prototyping, designing, and building an FRC robot
- Expedited design time by 38% compared to previous years by establishing new design standards and practices for the team
- Redesigned a key sub-system after observing a 68% malfunction rate from collisions at the initial competition, reducing breakdowns by 95%
- Implemented a part tracking system that eliminated unnecessary duplicated parts
- Reached Semifinals at the Ontario Provincial Championship and qualified for FIRST World Championship

Co-Founder / 11230A Mechanical Lead

2025 – 2026

Iroquois Ridge High School VEX Robotics Teams 11230A, 11230B, 11230C

Oakville, ON

- Founded a 3-team VEX Robotics V5 program at IRHS, mentoring students on teams 11230B and 11230C through their first robot builds, earning 16th and 18th competition finishes respectively
- Designed and built the 11230A competition robot using limited classroom kits and a \$600 budget, qualifying for the VEX Ontario Provincial Championship in the team's rookie year
- Built a Google Sheets inventory tracker to monitor existing parts and identify part shortages across all 3 teams, streamlining purchasing and budget allocation

PROJECTS

Robotic Hand | *Fusion 360, 3D Printing, DFA*

Jan. 2026

- Engineered an 11-DOF robotic hand with independent finger ab-adduction and actuation and wrist rotation, using a dual servo tendon-driven system per finger controlled via a Raspberry Pi Pico
- Validated functional grip strength by holding up a 250g ball, meeting the project's core deliverable
- Improved hand maintenance time by 80%, by iterating on palm cover design and simplified tendon layouts

Differential Swerve Drive | *Fusion 360, 3D Printing, Python, Technical Documentation*

Nov. 2025

- Created a differential swerve drivetrain using 3D-printed components and 4 VEX V5 11W motors, integrating steering and drive into a dual-motor differential mechanism to exceed the 6.5ft/s target, reaching 7.0ft/s
- Programmed the swerve drive control system in Python, translating joystick input into real-time motor commands
- Documented the entire engineering process within a 25-page [design report](#)

Mechanical Fanning Wallet | *Onshape, 3D Printing, DFM*

Nov. 2025

- Designed a slim form factor fanning wallet with an 8 card capacity
- Refined the design through 13 3D-printed prototypes, achieving 99% card-fanning consistency from over 100 trials, then developed production-ready CNC manufacturing drawings adhering to JLCCNC's guidelines

Cycloidal Gearbox | *Onshape, 3D Printing, Data Acquisition*

Jul. 2025

- Engineered a 31:1 cycloidal gearbox rated for 0.45Nm max input torque in Onshape
- Slashed backlash by 94%, from 7° to 0.4°, compared to the team 1360's 2025 robot arm, by replacing the traditional gearbox with the cycloidal gearbox
- Refined mechanical efficiency from 64% – 78%, measured with a digital torque wrench, by enhancing lubrication

SKILLS

Software: SolidWorks, Fusion 360, Onshape, Excel

Manufacturing: 3D Printer, CNC Mill, CNC Router, Laser Cutter, Vertical Mill, Lathe

Technical Skills: DFM/DFA, GD&T, Technical Documentation, Python